

SpaceX Falcon 9 First Stage Landing Prediction

IBM Applied Data Science Capstone Project Report

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1. Executive Summary

The primary objective of this capstone project is to develop a predictive machine learning pipeline that determines whether the Falcon 9 first stage booster will land successfully during commercial launches. Recovering and reusing the first-stage booster saves SpaceX over 60% in manufacturing costs, enabling launch bidding prices of ~\$62M compared to competitors' expendable launches costing ~\$165M+. Successful recovery prediction is a key asset for competitors and insurers bidding and pricing space logistics services.

This project uses SpaceX REST API records and scraped Wikipedia tables to compile a 90-flight history. EDA indicated launch successes increased over time as technology matured. GIS mapping via Folium verified launch pads are positioned near coastlines (<1km) for maritime recovery trajectories. SQL auditing with SQLite examined historical parameters. Finally, machine learning classification pipelines were constructed using Support Vector Machines, Logistic Regression, Decision Trees, and K-Nearest Neighbors, optimized via GridSearchCV. Logistic Regression, SVM, and KNN tied with the highest test accuracy of 83.33% on the hold-out test set.

2. Introduction

SpaceX (Space Exploration Technologies Corp.) has pioneered reusable rocket technologies. While traditional space transportation discard boosters, SpaceX Falcon 9 recovers the first stage by returning it vertically onto ground pads or autonomous spaceport drone ships. Rocket hardware is the single largest capital expense, making first stage recovery the main driver behind SpaceX's cost leadership.

This report documents a structured data science workflow to build a landing prediction model based on public flight history variables, helping external stakeholders quantify operational and financial risks associated with launch profiles.

3. Data Collection

Data collection was split into API ingestion and web scraping:

SpaceX API Ingestion: Queries live launch logs. Since relational details are represented as UUID strings, auxiliary queries resolved nested IDs to retrieve booster versions, launchpad coordinates, core landing pad names, and payload mass metrics.

Web Scraping: Scraped Wikipedia tables of Falcon 9 flights to cross-reference dates, customers, and landing states, establishing a clean dataset covering 90 Falcon 9 flights.

4. Data Wrangling

Cleaning and feature engineering were performed on the raw flights dataset:

Filtering: Removed early Falcon 1 launches to focus on Falcon 9 launches.

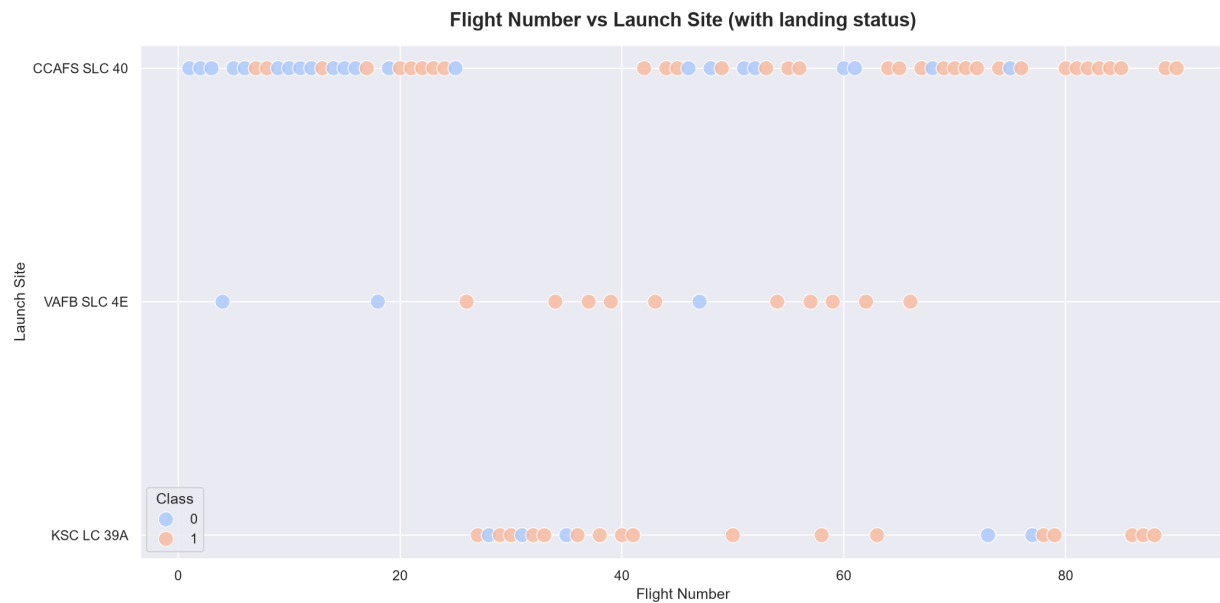
Imputation: Identified null entries in payload mass and imputed them using the column mean of 6,104.96 kg.

Target Labeling: Transformed descriptive outcome strings into binary class variables (1 for successful recoveries, 0 for failures/unrecovered launches).

5. Exploratory Data Analysis (EDA)

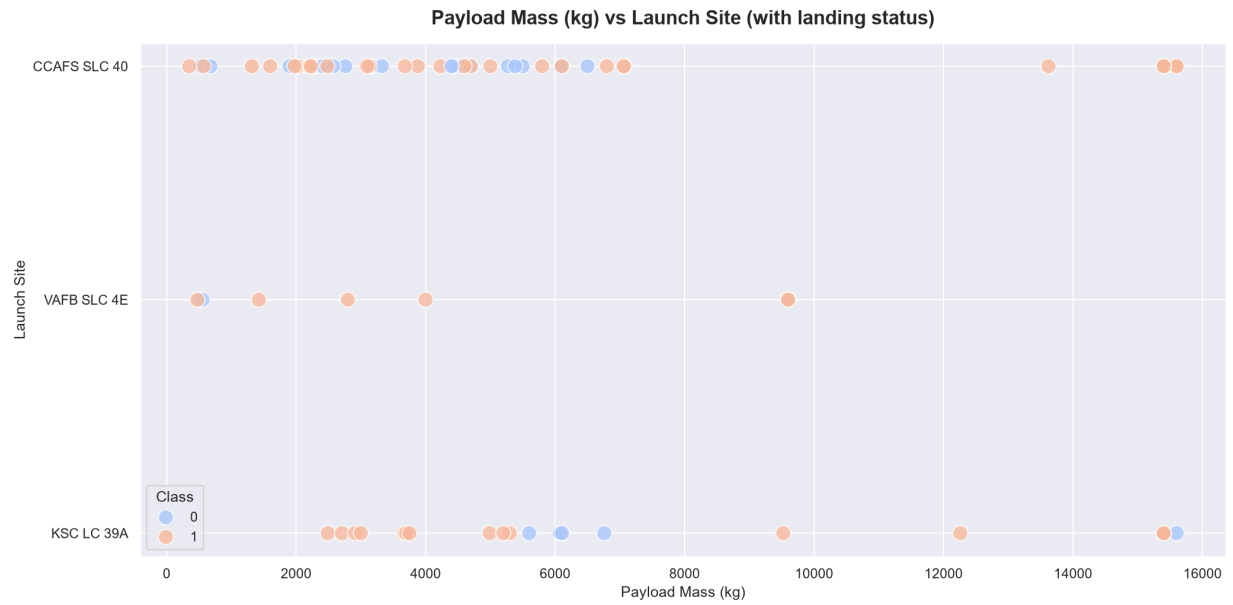
Visual patterns in the cleaned dataset were evaluated using Seaborn and Matplotlib plots:

5.1 Flight Number vs Launch Site



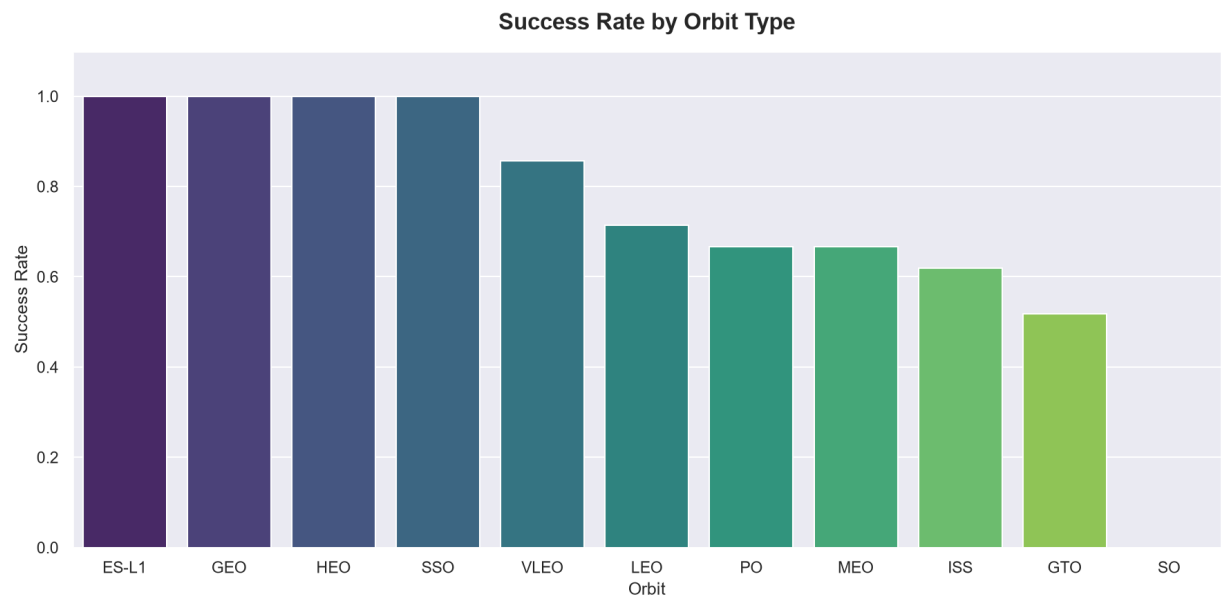
Visualizes launch pad utilization over flight history. Early flights show a concentration of failures (orange dots) mostly at CCAFS LC-40. Later flights show a dominant success rate (blue dots) across all sites, validating technological maturity.

5.2 Payload Mass vs Launch Site



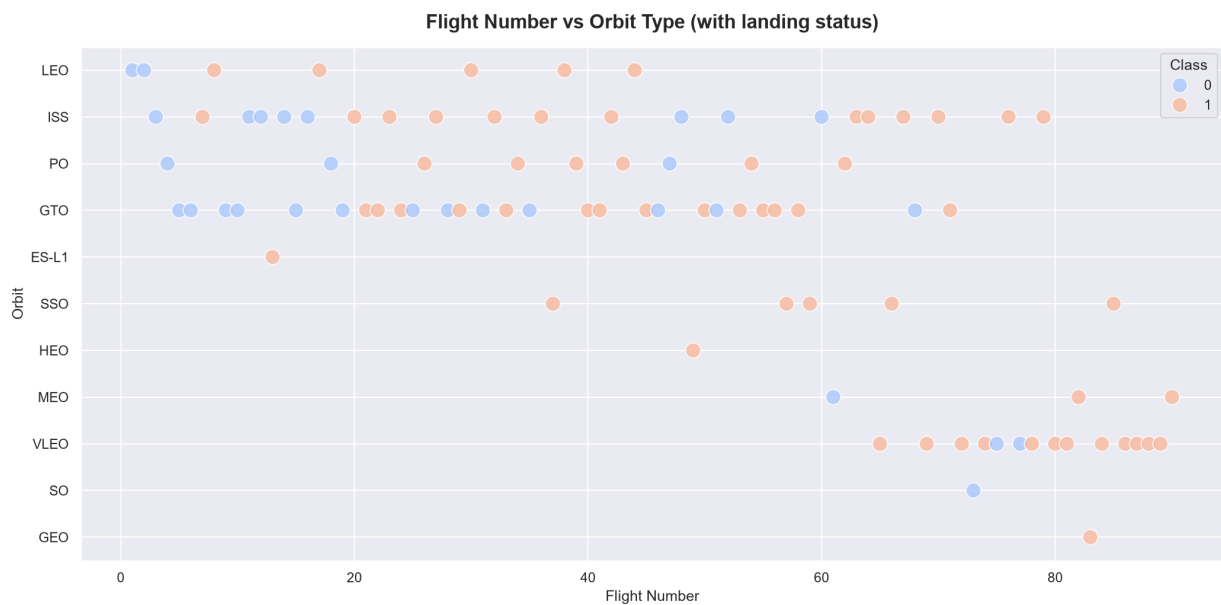
Shows that heavy payloads exceeding 10,000 kg are launched from KSC LC-39A and CCAFS LC-40. VAFB SLC-4E handles payloads below 10,000 kg. High-mass payloads maintain a strong success record on later flights.

5.3 Success Rate by Orbit Type



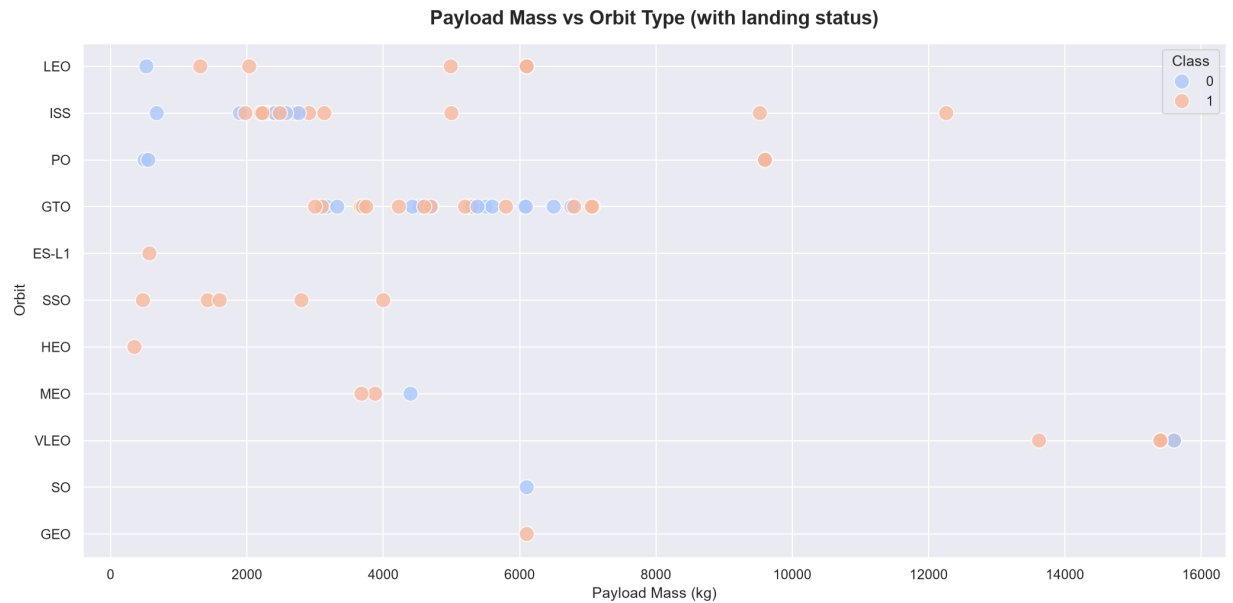
Orbits ES-L1, GEO, HEO, and SSO have a perfect 100% success rate. High-energy Geostationary Transfer Orbits (GTO) show lower success rates (~50%) due to higher entry velocities and lower landing fuel margins.

5.4 Flight Number vs Orbit Type



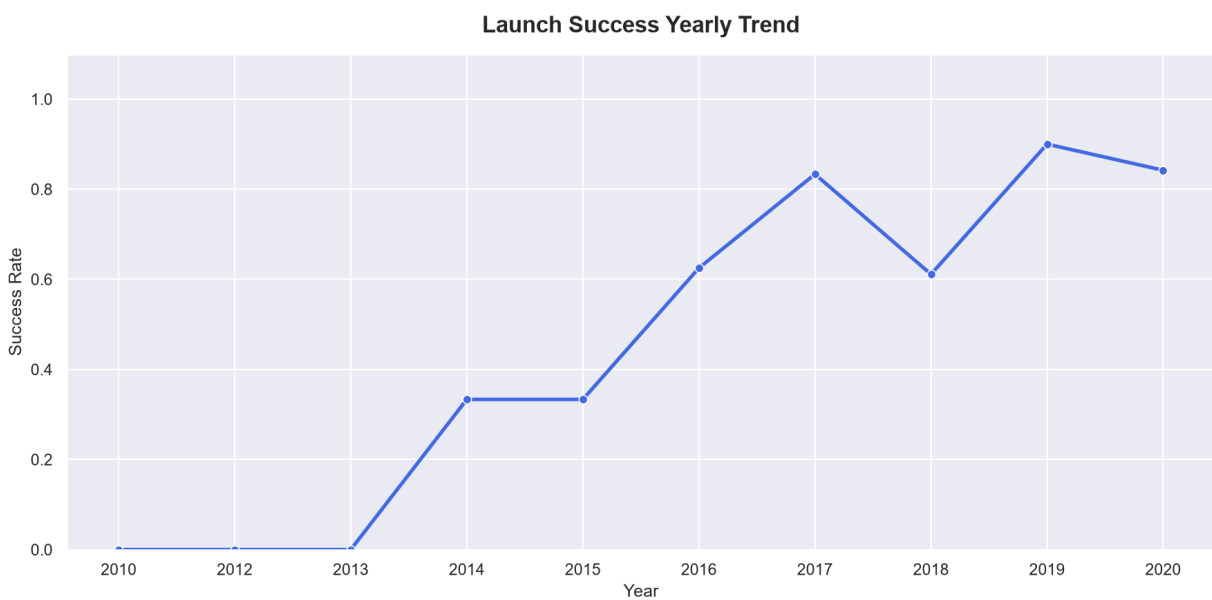
Demonstrates orbit destination shifts over flight sequence. Early flights targeted LEO/ISS cargo runs, while later flights expanded to high GTO and Polar orbits.

5.5 Payload Mass vs Orbit Type



Polar Orbits (PO) have payload weights clustered near 9,600 kg. LEO shows the widest range, including Starlink launches up to 15,600 kg.

5.6 Success Rate Yearly Trend



The line chart shows launch recovery success rising rapidly from 0% in 2013 to over 80% in 2020, representing the learning curve of SpaceX engineering.

6. SQL Analysis

A SQLite database was constructed and queried to audit launch performance metrics:

Launch Site Listing: Identified unique launch sites (CCAFS, KSC, VAFB).

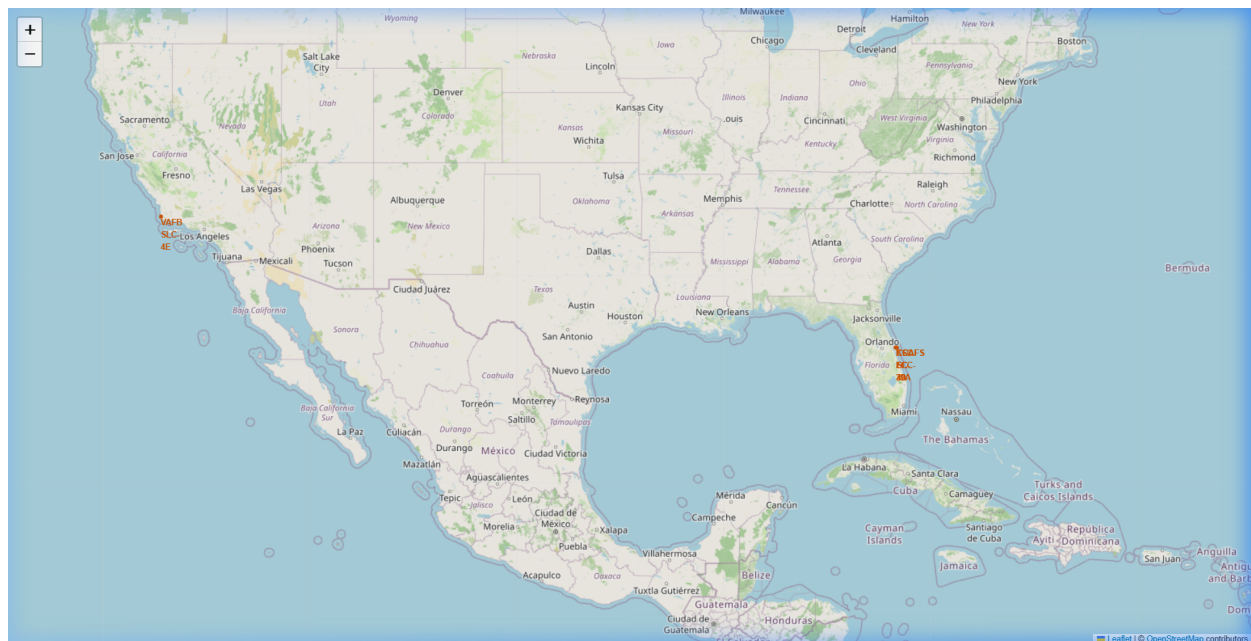
NASA ISS Contracts: Calculated that a total of 45,596 kg was delivered under NASA CRS cargo contracts.

Average Payload: Identified booster version F9 v1.1 average payload is 2,928.4 kg.

Historical Milestones: Determined the first successful ground pad landing occurred on 2015-12-22.

7. Folium Map Analysis

Interactive spatial GIS maps were constructed using Folium to evaluate pad layout heuristics:



Launch pads are positioned within 1 km of the shoreline to allow safe overwater flight trajectories. They are located close (<1.3 km) to highways and railways to facilitate transportation logistics, while keeping a safe distance (>15 km) from close cities to minimize noise and shockwave hazards.

8. Machine Learning Prediction

The features were standardized via StandardScaler and trained on an 80/20 train-test split. Hyperparameters were tuned using GridSearchCV:

Logistic Regression: Tuned regularization parameter C. Best C = 0.01. Test Accuracy: 83.33%.

Support Vector Machine (SVM): Tuned kernel types and regularization. Best C = 1.0, gamma = 0.0316, kernel = 'sigmoid'. Test Accuracy: 83.33%.

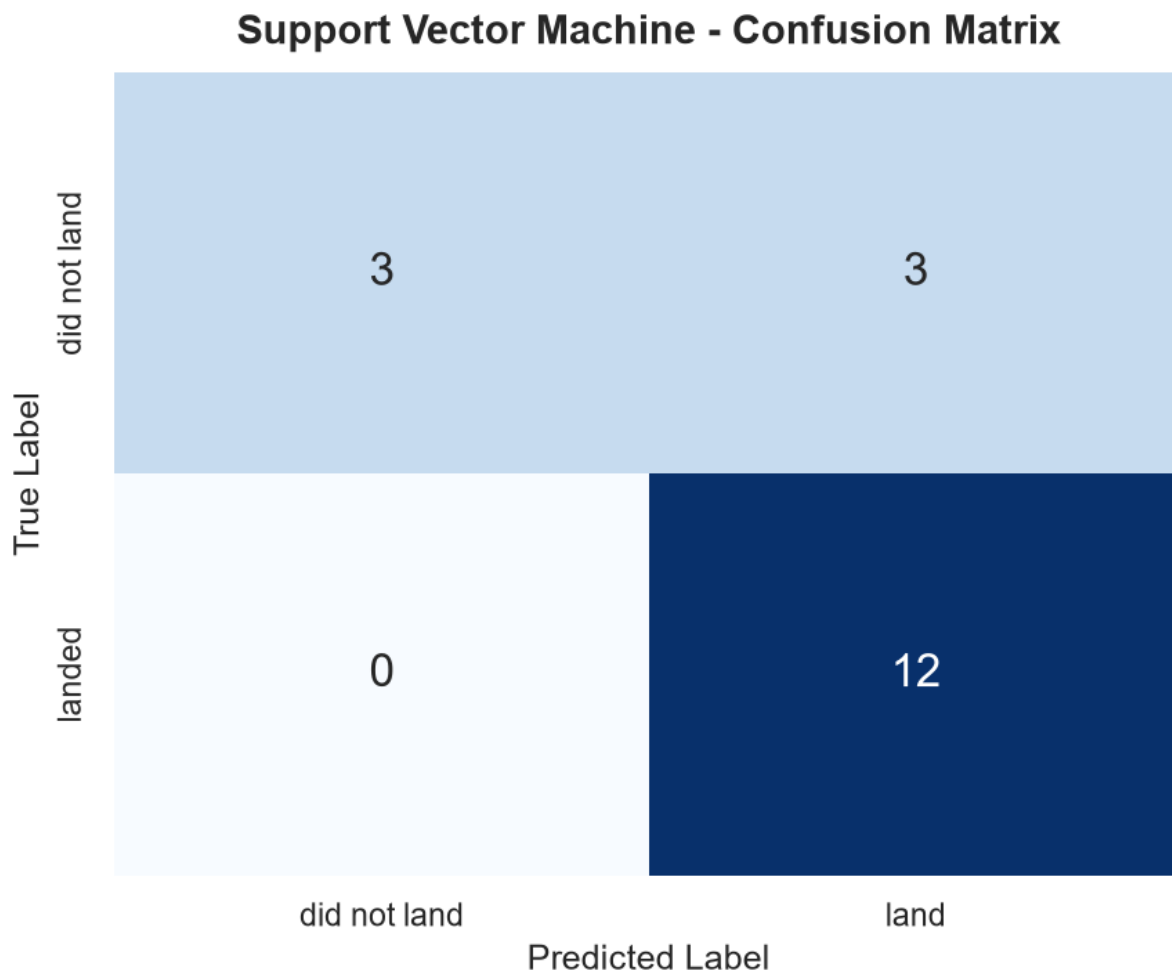
Decision Tree Classifier: Tuned splitting depth and thresholds. Best criterion = 'gini', depth = 6. Test Accuracy: 77.78%.

K-Nearest Neighbors (KNN): Tuned neighbor size and algorithm. Best neighbors = 10, distance p = 1. Test Accuracy: 83.33%.

8.1 Classifier Performance Comparison

Method	Validation Accuracy (CV)	Test Accuracy
Support Vector Machine (SVM)	84.82%	83.33%
Logistic Regression	84.64%	83.33%
K-Nearest Neighbors (KNN)	84.82%	83.33%
Decision Tree	90.54%	77.78%

8.2 Confusion Matrix Reference



LR, SVM, and KNN classifiers tied with the highest test accuracy of 83.33%. SVM is recommended as the final predictor due to its optimal margin boundaries that prevent overfitting on sparse features.

9. Results & Key Insights

1. Landing success is strongly correlated with booster blocks and flight numbers, showing high reliability (80%+) after 2016.
2. Orbits like ES-L1, GEO, HEO, and SSO have 100% successful landings. High-energy GTO orbits show ~50% success rates.

3. Spacecraft recovery is a crucial business advantage, offering up to \$50M in manufacturing cost savings per flight.

10. Conclusion and Recommendations

This project successfully demonstrated that SpaceX booster recovery can be predicted with an accuracy of 83.33% using SVM and Logistic Regression. Competitors can utilize these predictions to evaluate launch pricing structures, and insurers can adjust GTO launch premium risks dynamically.

11. References

1. IBM Data Science Capstone Course Materials, 2026.
2. SpaceX API Documentation, <https://github.com/r-spacex/SpaceX-API>
3. Pedregosa et al., Scikit-learn: Machine Learning in Python, JMLR, 2011.
4. Wikipedia: List of Falcon 9 launches,
https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches